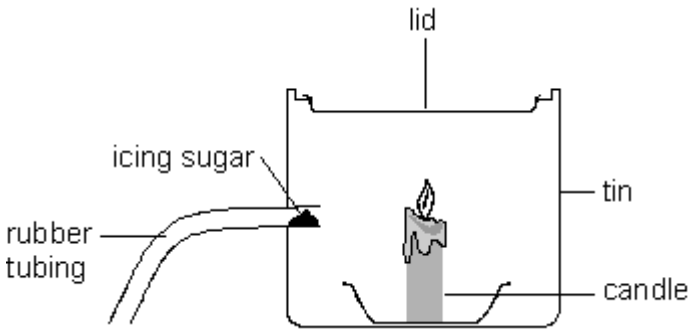
Q1.

A teacher set up the following apparatus behind a safety screen.
She placed 1 g of icing sugar in the end of the rubber tubing inside the tin, as shown below.



The teacher blew through the other end of the rubber tubing.
The icing sugar came into contact with the flame.
There was a loud explosion and the lid was blown off the tin.

(a) Complete the following sentence describing the energy changes which took place.

..... energy in the icing sugar changed to

..... energy and ..... energy.

3 marks

(b) As a result of the explosion, the lid of the tin was pushed off.
Explain what had happened to the gas molecules inside the tin to make this happen.

.....

.....

.....

.....

2 marks

(c) When icing sugar is burned in this experiment, the gas **used** and the gas **produced** are the same as when energy is released from sugar in the cells of the body.

(i) Which gas, in the air, is **used** when the icing sugar burns?

.....

1 mark

(ii) Give the name of the gas **produced** when the icing sugar burns.

.....

1 mark

(d) The table below shows the energy values of four food substances.

food substance	energy value, in kJ per 100 g
icing sugar	1680
curry powder	979
flour	1450
custard powder	630

The teacher repeated the experiment with 1 g of custard powder.
What difference would this make to the experiment?

.....

.....

1 mark
Maximum 8 marks

Q1.

(a) chemical

*accept 'potential' **or** 'stored'*

1 (L6)

any **two** from

- sound
- thermal

  accept 'heat'
- kinetic

  accept 'movement'
- light

2 (L6)

(b) any **two** from

- they gained energy

  accept 'they move more quickly'
- they hit the lid with greater force

  accept 'they hit the lid harder'
- they hit the lid more often

  accept 'the pressure inside the tin increased'

  accept 'the molecules are closer together'

  accept 'more molecules are present'

2 (L6)

(c) (i) oxygen

accept 'O_2'

1 (L6)

(ii) any **one** from

- carbon dioxide

  accept 'CO_2'
- water vapour

  accept 'H_2O'

  accept 'carbon monoxide'

1 (L6)

(d) any **one** from

- it was quieter
- the lid didn't move as high

accept 'the lid was not pushed off'

- less energy released

  accept 'it does not work'

1 (L5)

[8]